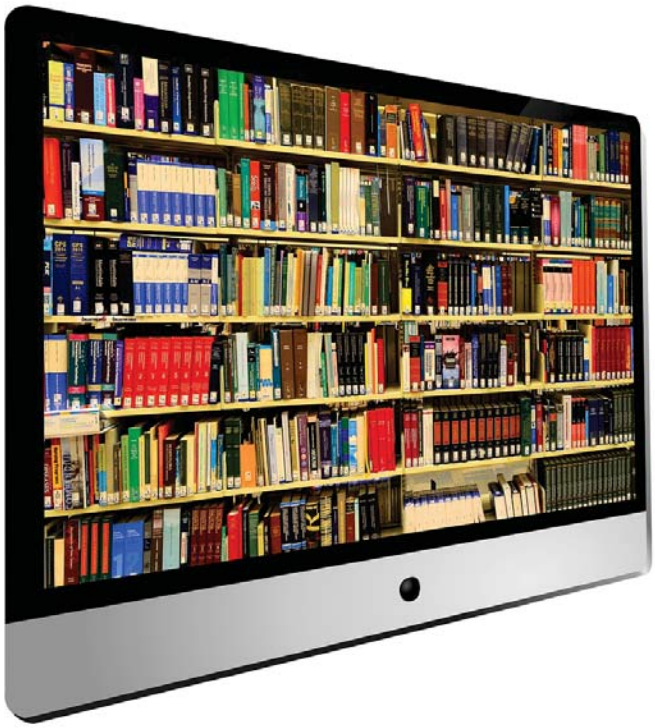
# THE VIRTUAL MEET EXPERIENCE

## 2019 - 2020

# HS VIRTUAL CHALLENGE MEET #4



# COMPUTER SCIENCE

**DO NOT OPEN TEST UNTIL TOLD TO DO SO**

# 2019-2020 Virtual Challenge Meet #4 – Computer Science

Note: Correct responses are based on **Java SE Development Kit 12 (JDK 12)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 12 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: import static java.lang.System.*;**

**QUESTION 1**

Which of the following is equivalent to the expression `0b100101 + 0xac`?

A. $1320_5$ B. $21202_3$ C. $255_9$ D. $320_8$ E. $182_{11}$

**QUESTION 2**

What is output by the code to the right?

A. `123` B. `0`
C. `165`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
out.println(15 * 8+3);
```

**QUESTION 3**

What is output by the code to the right?

A. `hi dude!`
B. `Hi dude!`
C. `HI DUDE!`
D. `HI dude!`
E. There is no output due to an error.

```
String s="hi";
out.printf("%S dude!%n",s);
```

**QUESTION 4**

What is output by the code to the right?

A. `1` B. `22`
C. `-1` D. `-22`
E. There is no output due to an error.

```
String a="Gotcha";
String b="Gotyou";
int c=a.compareTo(b);
out.println(c);
```

**QUESTION 5**

What is output by the code to the right?

A. `True` B. `False`
C. `false` D. `true`
E. There is no output due to a syntax error.

```
boolean a=true;
boolean b=a && !a;
boolean c= a && (b || a);
out.println(c);
```

**QUESTION 6**

What is output by the code to the right?

A. `6.0` B. `5.0`
C. `7.0` D. `6`
E. There is no output due to an error.

```
int y=6;
double h=Math.ceil(y);
out.println(y);
```

**QUESTION 7**

What is the output by the code to the right?

A. `35` B. `3`
C. `15` D. `1`
E. `5`

```
int y=6;
if(y<5)
 if(y>3)out.print(1);
 else out.print(2);
else
 if(y<8)out.print(3);
 else out.print(4);
out.println(5);
```

**QUESTION 8**

What is the output by the code to the right?

A. 162 B. 163

C. 56 D. 55

E. 487

```
int h=3;
for(int y=0;y<4;y++)
 h=(h*3)-2;
out.println(h);
```

**QUESTION 9**

What is output by the code to the right?

A. ht elleeHor

B. e HatheraeanHHaeblelo

C. nhaaeeaebHo

D. aeeaebHoHth

E. haaeeaebHoH

```
String s="Hello there";
String t="General Kenobi";
for(int h=0;h<10;h++) {
 char c1=s.charAt(h);
 char c2=t.charAt(h+2);
 s=s.substring(h+1)+c2
   +s.substring(0,h);
 t=t.substring(h+2)+c1
   +t.substring(0,h+1);
}
out.println(s);
```

**QUESTION 10**

What is the output by the code to the right ?

A. 3 B. 8

C. There is no output due to a syntax error.

D. There is no output due to a runtime error

E. Output cannot be determined until runtime.

```
int[] a= {1, 2, 3, 4, 5};
a[1]=a[2]+a[4];
out.println(a[2]);
```

**QUESTION 11**

What is the output by the code to the right?

A. 1 B. 3

C. 6 D. 10

E. There is no output due to a runtime error.

```
Scanner s=new Scanner("1 2 3 4");
int y=0;
while(s.hasNextInt())
 y+=s.nextInt();
out.println(y);
```

**QUESTION 12**

What is the output by the code to the right ?

A. 210 B. 232

C. 256 D. 189

E. 0

```
int h=0;
for(int y=0;y<21;y++)
 h+=y;
out.println(h);
```

**QUESTION 13**

What is the correct order of precedence for the operators to the right ?

A. III, II, I, IV B. II, III, I, IV

C. I, III, II, IV D. III, II, IV, I

E. IV, III, II, I

```
I. ++(post-increment)
II. +(addition)
III. +(unary)
IV. &(bitwise)
```

**QUESTION 14**

What is the output by the code to the right ?

A. 4 B. 16

C. 8 D. 64

E. 32

```
out.println(Float.SIZE);
```

**QUESTION 15**

What is the output by the code to the right ?

A. 423

B. 212

C. 17

D. 19

E. There is no output due to an error.

```
ArrayList<Integer> a=new
ArrayList<Integer>();
a.add(4);
a.add(9);
a.add(10);
a.add(212);
a.add(17);
a.replaceAll(n -> n*2-1);
a.remove(19);
out.println(a.get(1));
```

**QUESTION 16**

What is the output by the code to the right ?

A. Equal? false B. Equal? true

C. true D. false

E. There is no output due to a syntax error.

```
String a="Hello";
String b=a;
out.println("Equal? " + a==b);
```

**QUESTION 17**

What is the output by the code to the right ?

A. -17.0 B. 2

C. -17 D.2.0

E. There is no output due to an error.

```
int y=17;
double d=Math.copySign(y, -2);
out.println(d);
```

**QUESTION 18**

What is output by the code to the right

A. SO*eb*dy *ce t*d *e

B. SO*eb*dy **ce t**d *e

C. SOmebody once told me

D. There is no output due to a syntax error.

E. There is no output due to a runtime error.

```
String h="SOmebody once told me";
h=h.replaceAll("[g-q]+", "*");
out.println(h);
```

**QUESTION 19**

What could not replace **<1*>** in the code to the right, so that the code will compile?

A. char a='a'; B. int a=56;

C. short a=87; D. long a=34;

E. More than one of the above.

```
<1*>;
switch(a) {
 //implementation
}
```

**QUESTION 20**

What is $64_{10}$ - $11111_2$ ?

A. $32_{10}$ B. $100001_2$ C. $31_8$ D. $22_{16}$ E. None are equivalent

**QUESTION 21**

What output by the code to the right ?

A. 1056 B. 1060 C. 1057 D. 1058 E. 1059

```
String s=Integer.toString(81,5);
int y=Integer.parseInt(s,6);
s=Integer.toBinaryString(y);
y=Integer.parseInt(s,3);
System.out.print(y);
```

**QUESTION 22**

What is the output by `//q22` in the client code to the right?

A. `7`

B. `Sam`

C. `5`

D. `Sam null`

E. There is no output due to an error.

**QUESTION 23**

What is the output by `//q23` in the client code to the right?

A. `Sam 7`

B. `null 0`

C. `Sam 7: Sam, null; 5`

D. `Sam 5`

E. There is no output due to an error.

**QUESTION 24**

What is the output by `//q24` in the client code to the right?

A. `212`

B. `5`

C. `Sam Johnson`

D. `Sam Johnson 212: Sam, Johnson; 5`

E. This is no output due to an error.

**QUESTION 25**

What is the output by `//q25` in the client code to the right?

A. `Sam Johnson 212: Sam, Johnson; 212`

B. `Sam Johnson 5: Sam, Johnson; 5`

C. `Sam Johnson 212: Sam, Johnson; 5`

D. Output cannot be determined until runtime

E. This is no output due to a runtime error.

```
class A{
 private int y;
 String name;
 public A(String n, int h) {
  y=h;
  name=n;
 }
 public int getY() {
  return y;
 }
 public String toString() {
  return name+" "+y;
 }
}
class B extends A{
 private int y;
 String fname, lname;
public B(String fn, String ln, int h)
{
  super(fn+" "+ln, h);
  y=5;
  fname=fn;
  lname=ln;
 }
 public String toString() {
return super.toString()+": "+fname+",
      "+lname+"; "+y;
 }
}

/////////////client code/////////////
A a=new A("Sam", 7);
out.println(a.getY());    //q22

out.println(a);    //q23

B b=new B("Sam", "Johnson", 212);
out.println(b.getY());    //q24

out.println(b);    //q25
```

**QUESTION 26**

What is output by the code to the right?

A. `9 10 11 12 13 14 15 16 17 9`

B. `1 2 3 4 5 6 7 8 9 10`

C. `8 9 10 11 12 13 14 15 16 17`

D. `9 10 11 12 13 14 15 16 17 18`

E. There is no output due to an exception.

```
int [] bob = new int [10];
for(int i = 0; i < bob.length; i++) {
 bob[i] = i;
 for(int j = i + 1;
              j < bob.length; j++) {
  bob[j] = bob[i];
  bob[i] = j + i;
 }
}
for(int i = 0; i < bob.length; ) {
 out.print(bob[i++] + " ");
}
```

**QUESTION 27**

If the call

`rec("chickens", "chickens".length(), 's')`

was made, what would be returned?

A. `ssschickens`
B. `ssschicken`
C. `chickens`
D. `chickenssss`
E. There is no output due to a runtime error.

```
public String rec(String s, int i, char c) {
 if(i> 10) {
  return s;
 }
 return rec(s, i + 1, c) + c;
}
```

**QUESTION 28**

What is output by the code to the right?

A. 25
B. 17
C. 95
D. 34
E. 21

```
public int rec(int t, int o) {
 if(t % o == 0) {
  return t;
 }
 if(t > o) {
  return o + rec(t/o, o);
 }
 return t + rec(t, o/t);
}


////////////////////////////////////////
/////////////CLIENT CODE/////////////
out.println(rec(74, 17));
```

**QUESTION 29**

What is equivalent to the expression `A&&B`?

A. `A||B`
B. `!A && B`
C. `!A || !B`
D. `!(!A || !B)`
E. `!(!A && !B)`

**QUESTION 30**

What of the following is demonstrated by method `go`?

A. Pythagorean Theorem
B. Dynamic Programming
C. Dijkstra's algorithm
D. A logarithmic sequence
E. The Fibonacci Sequence

```
public int v(int val, int b) {
 int i = 1;
 while(val > b) {
    val = val/b;
    i++;
 }
 return i;
}
```

**QUESTION 31**

What is output by the code at right?

A. `8 77 45 44 8 45`
B. `8 44 45 77`
C. `8 8 44 45 45 77`
D. There is no output due to a syntax error.
E. There is no output due to a NullPointerException.

```
TreeSet<Integer> i;
i = new TreeSet<Integer>();

ArrayList<Integer> ar;
ar = new ArrayList<Integer>();

ar.add(8);
ar.add(77);
ar.add(45);
ar.add(44);
ar.add(8);
ar.add(45);
i.addAll(ar);

Iterator it = i.iterator();

while(it.hasNext()) {
  out.print(it.next() + " ");
}
```

**QUESTION 32**

What could replace **`<1*>`** in the code to the right, so that the code works as intended?

A. `0`

B. `3`

C. `1`

D. More than one of the above.

E. None of the above.

**QUESTION 33**

Assuming the code above works as intended, what is output by the code to the right ?

A. `Shrek Wes Sheck West Sec West Check Wes Shark Wash`

B. `Shrek Wes Sheck West Sec West Check Wes Sheck Wasp Shark Wash`

C. `Shrek Wes Sheck West Check Wes Shark Wash`

D. `Shrek Wes Sheck West Sec West Shark Wash`

E. There is no output due to a runtime error.

```
Stack<String> s = new Stack<String>();
s.add("Shrek Wes");
s.push("Sheck West");
s.push("Sec West");
s.add("Check Wes");
s.add("Sheck Wasp");
s.pop();
s.pop(<1*>);
s.push("Shark Wash");

while(!s.isEmpty())
{
    out.print(s.remove(0) + " ");
}
```

**QUESTION 34**

What could replace **<1*>** in the code to the right, so that `struct` can store any class, custom or otherwise?

A. `E`
B. `Object`
C. Nothing is needed to replace **<1*>**.
D. More than one of the above.
E. None of the above.

**QUESTION 35**

What could replace **<2*>** in the code to the right, so that the toString method returns a string with no leading or trailing spaces?

A. `h.substring(0,h.length()-1)`
B. `h`
C. `h.trim()`
D. A and C
E. There is no output due to an error.

**QUESTION 36**

What is output by the line marked `//q36` in the client code to the right ?

A. `2`
B. `1`
C. `0`
D. `3`
E. There is no output due to an error.

**QUESTION 37**

What is output by the line marked `//q37` in the client code to the right ?

A. `null`
B. `Hello`
C. `WE out here`
D. `General`
E. There is no output due to an error.

**QUESTION 38**

What is output by the line marked `//q38` in the client code to the right ?

A. `Kenobi WE out here`
B. `There WE out here General Kenobi`
C. `Hello There, General Kenobi`
D. There is no output due to a compilation error.
E. There is no output due to a runtime error.

```
class Structure{
int size;
<1*>[] struct;
int[] struct2;
public Structure(int s) {
 struct = new Object[s];
 struct2 = new int[s];
 size=0;
}
int add(int y, Object o) {
 int in=0;
 while(y<struct2[in] && in<size)in++;
 struct[in]=o;
 struct2[in]=y;
 size++;
 return in;
}
Object remove(int h) {
 Object o=struct[h];
 for(int y=h;y<struct.length-1;y++) {
  struct[y]=struct[y+1];
  struct2[y]=struct2[y+1];
 }
 struct[struct.length-1]=null;
 struct2[struct.length-1]=0;
 return o;
}
public String toString() {
 String h="";
 for(int y=0;
y<struct.length&&struct[y]!=null;y++)
  h+=struct[y]+" ";
  return <2*>;
 }
}
//////client code//////////////////////
Structure s=new Structure(10);
s.add(5, "Hello");
out.println(s.add(1, "There")); //q36
s.add(10, "General");
s.add(212, "Kenobi");
s.add(2, "WE out here");
Object o=s.remove(2);
out.println(o); //q37
out.println(s); //q38
```

**QUESTION 39**

What is the fully simplified form of the following expression?

```
((A && B)&&(!A && B)&&(B && C)&&(!B && C)&&(!C && A)&&(C && A))||(A||!A)
```

**QUESTION 40**

What is the in order traversal if the following values were inserted into a binary tree ?

```
5, 6, 7, 8, 212, 17, 1, -9, 0, 32, 4
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

# The Virtual Challenge Meets

# Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
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19) _____
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24) _____
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26) _____
27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

# 2019-2020 Virtual Challenge Meet #4
# Computer Science - KEY

1) B
2) A
3) D
4) D
5) D
6) D
7) A
8) B
9) E
10) A

11) D
12) A
13) C
14) E
15) E
16) D
17) A
18) A
19) D
20) B

21) C
22) A
23) A
24) A
25) C
26) A
27) D
28) A
29) D
30) D

31) B
32) E
33) D
34) B
35) C
36) B
37) A
38) A
39) true
40) -9 0 1 4 5 6 7 8 17 32 212

**Note to Graders:**

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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